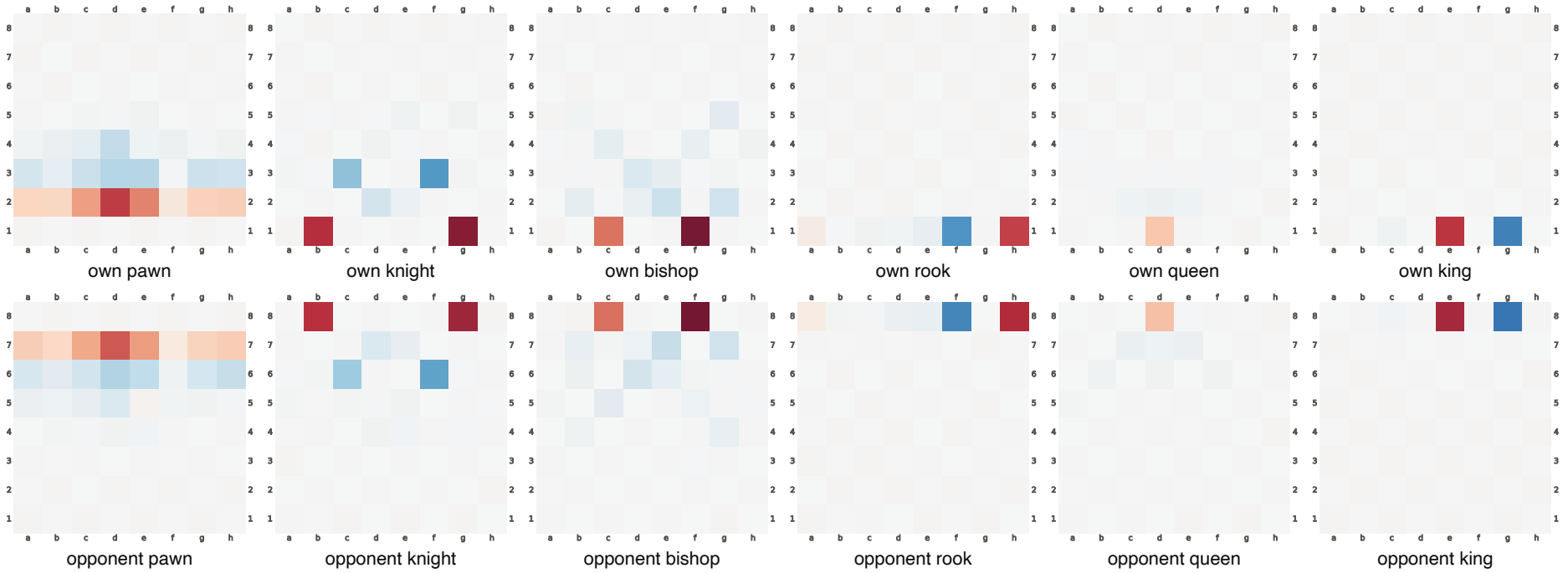
Maia-2 failures by game phase

Phase: opening

opening: 239456 non-forced positions | failure rate 0.555 | time_pressure mix {'>60s': 1.0, '30-60s': 0.0, '<30s': 0.0}

PC1 loadings, opening (14.5% of variance), side to move at the bottom

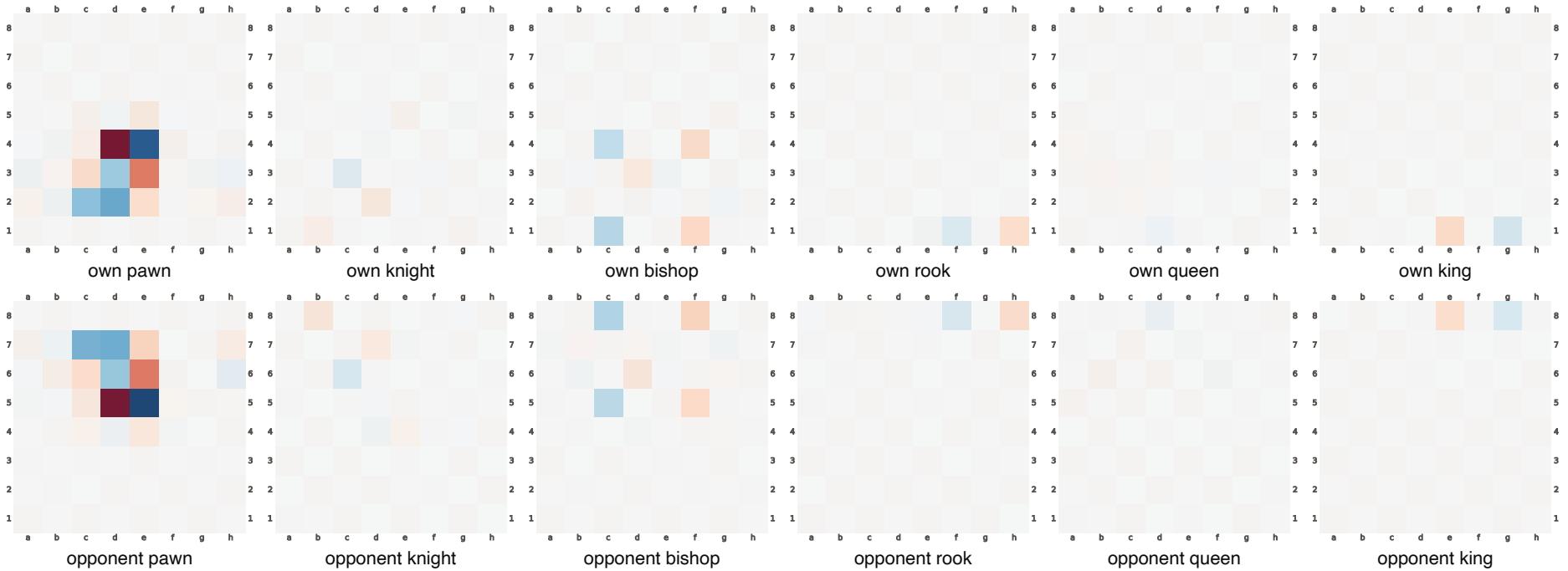


red = a piece on this square raises the PC score, blue = lowers it

-0.258 0.258

opening PC1 failure rate by score decile (low -> high): [0.51, 0.511, 0.484, 0.498, 0.486, 0.49, 0.524, 0.551, 0.613, 0.881]

PC2 loadings, opening (5.7% of variance), side to move at the bottom

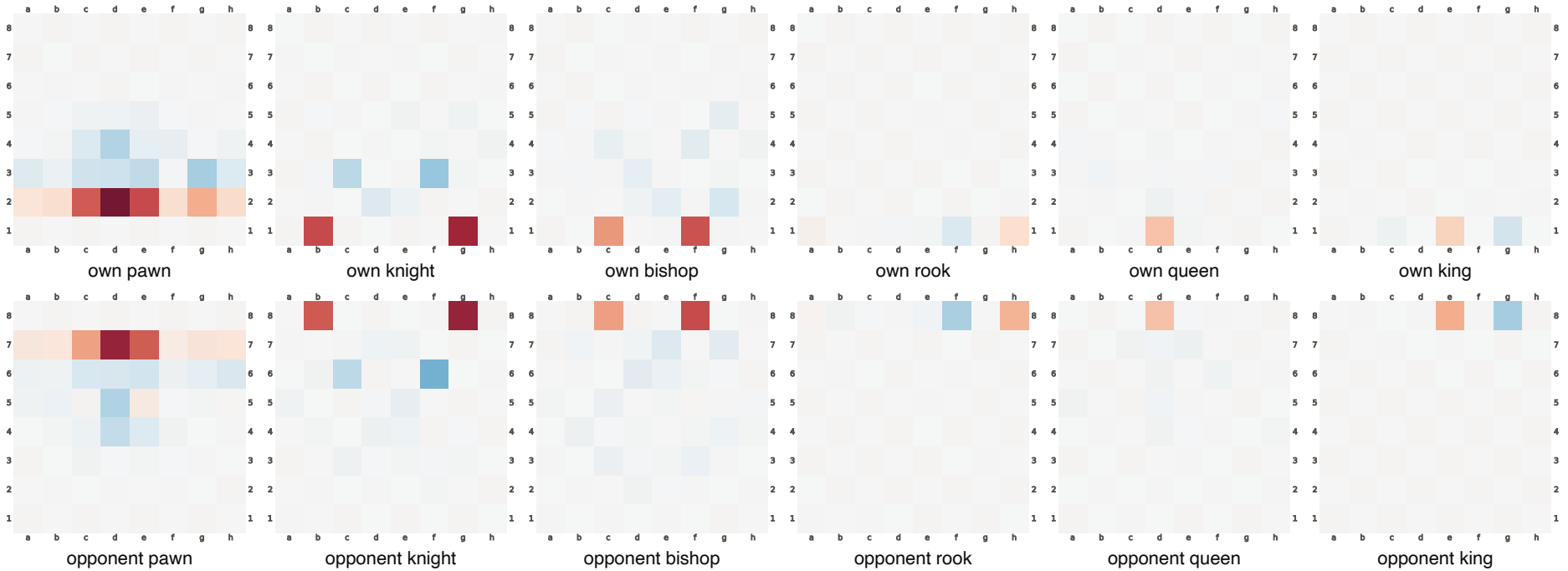


red = a piece on this square raises the PC score, blue = lowers it

-0.381 0.381

opening PC2 failure rate by score decile (low -> high): [0.564, 0.532, 0.624, 0.539, 0.7, 0.5, 0.541, 0.514, 0.526, 0.504]

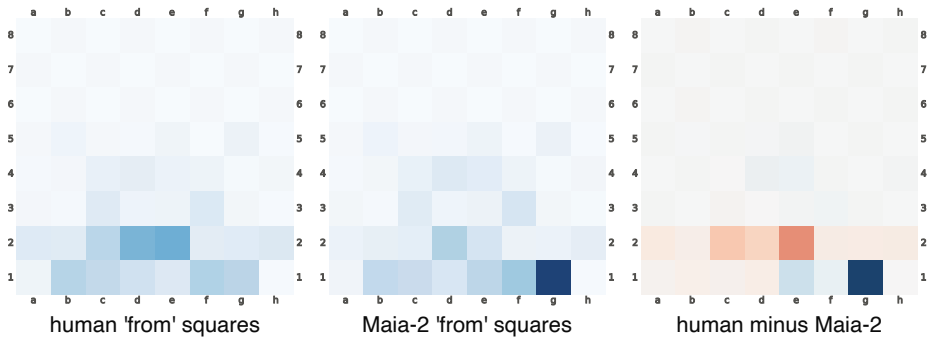
Where pieces stand when Maia-2 fails vs. succeeds (opening, n=20000)



percentage points: red = occupied more often in failed positions, blue = in correct ones

-13.9 13.9

Failed predictions, opening: 'from' squares (n=132841, mover at the bottom)



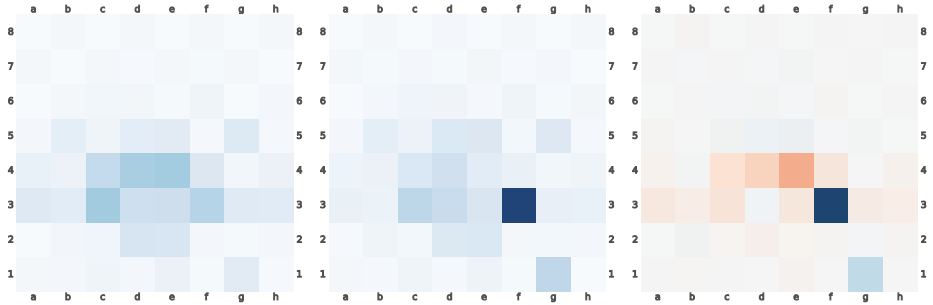
share of moves in %

0 22.5

red = humans use this square more, blue = Maia-2 expects it more (pp)

-15.8 15.8

Failed predictions, opening: 'to' squares (n=132841, mover at the bottom)



human 'to' squares | Maia-2 'to' squares | human minus Maia-2

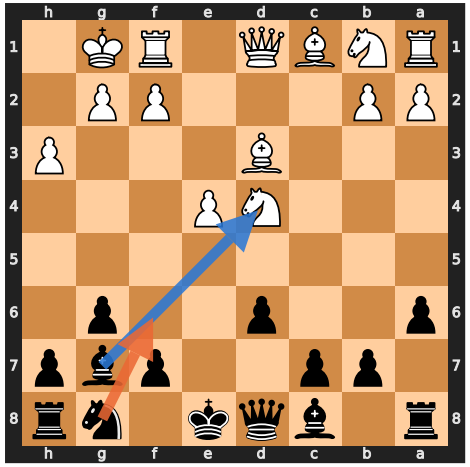
share of moves in %

0 — 23.4

red = humans use this square more, blue = Maia-2 expects it more (pp)

-15.8 — 15.8

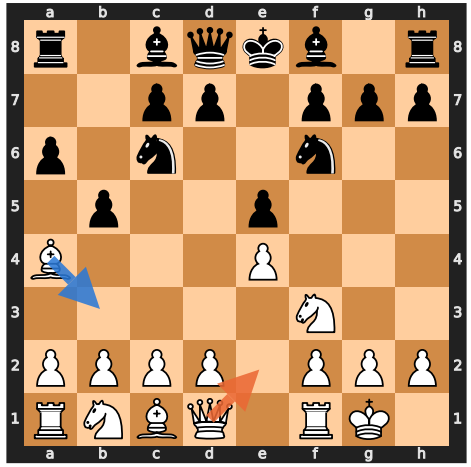
Most confident Maia-2 failures (opening)



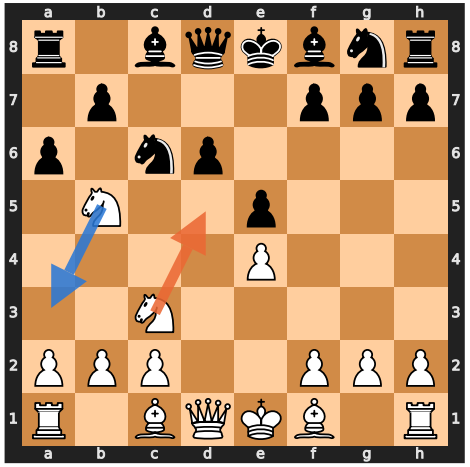
black to move, 2022 vs 1919, 707s left
Maia-2 g7d4 (100%) | human g8f6 (0%)
Pirc Defense



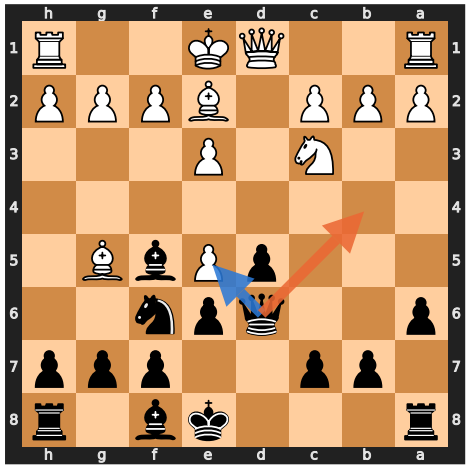
white to move, 1646 vs 1678, 586s left
Maia-2 a4b3 (100%) | human a4b5 (0%)
Four Knights Game: Spanish Variation



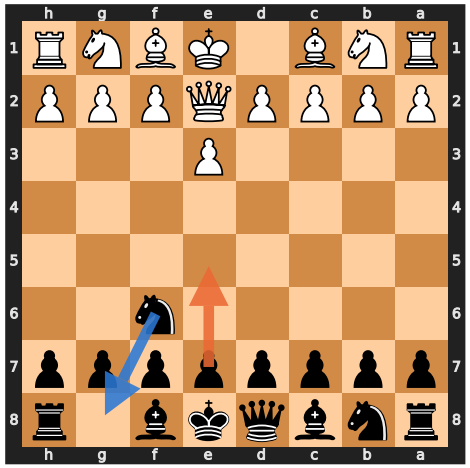
white to move, 2078 vs 2083, 177s left
Maia-2 a4b3 (100%) | human d1e2 (0%)
Ruy Lopez: Morphy Defense



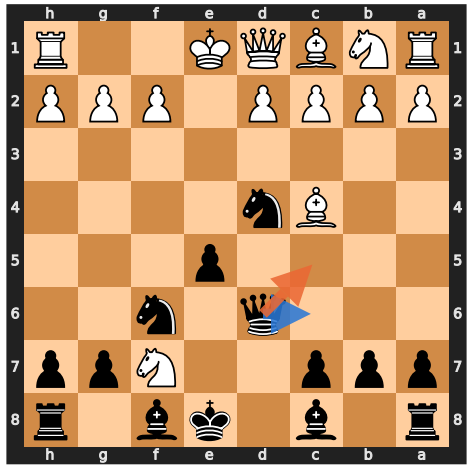
white to move, 2565 vs 2569, 175s left
Maia-2 b5a3 (100%) | human c3d5 (0%)
Sicilian Defense: Kalashnikov Variation



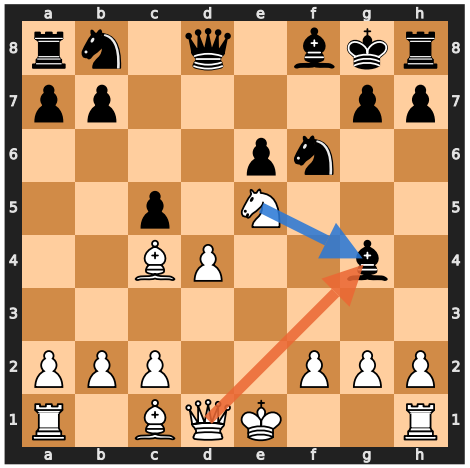
black to move, 2152 vs 2130, 151s left
Maia-2 d6e5 (100%) | human d6b4 (0%)
Queen's Pawn Game: Torre Attack



black to move, 2589 vs 2584, 180s left
Maia-2 f6g8 (100%) | human e7e5 (0%)
Van't Kruijs Opening



black to move, 1728 vs 890, 250s left
Maia-2 d6c6 (100%) | human d6c5 (0%)
Italian Game: Two Knights Defense, Fritz Variation



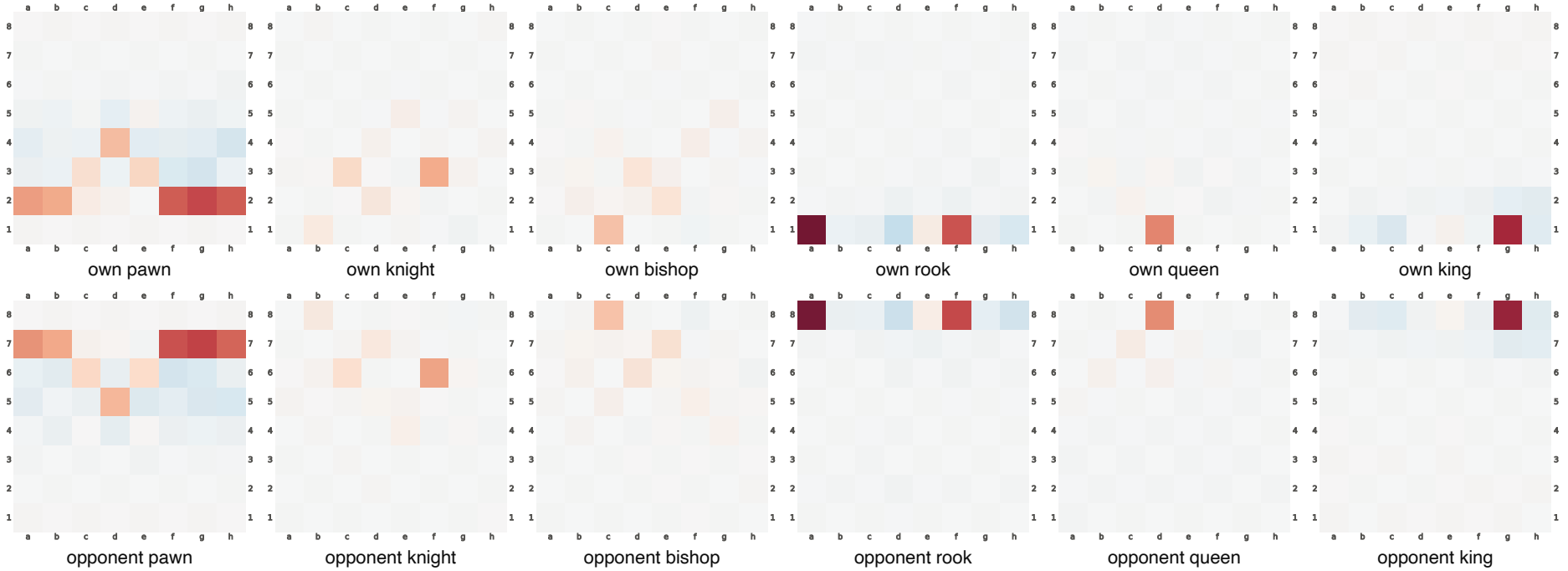
white to move, 2053 vs 1933, 169s left
Maia-2 e5g4 (100%) | human d1g4 (0%)
Caro-Kann Defense: Main Line

blue arrow = Maia-2 top prediction, orange arrow = move actually played

Phase: middlegame

middlegame: 279110 non-forced positions | failure rate 0.487 | time_pressure mix {'>60s': 0.89, '30-60s': 0.06, '<30s': 0.05}

PC1 loadings, middlegame (4.0% of variance), side to move at the bottom

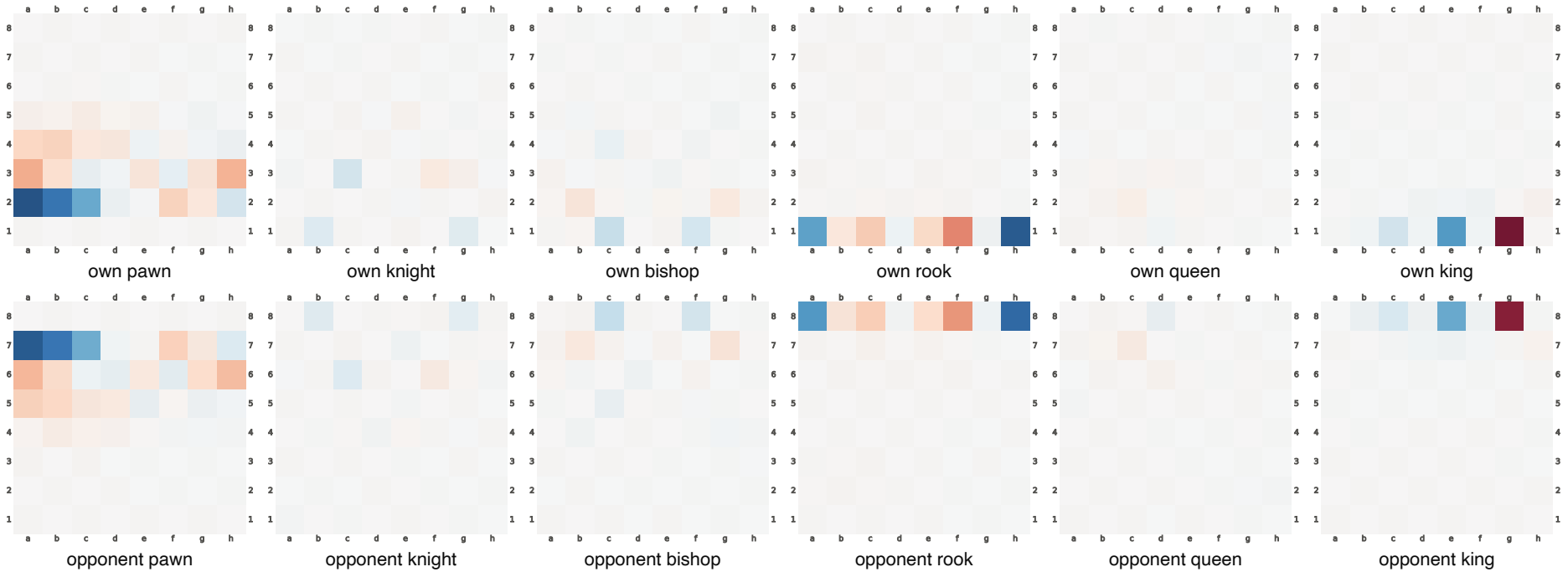


red = a piece on this square raises the PC score, blue = lowers it

-0.309 0.309

middlegame PC1 failure rate by score decile (low -> high): [0.49, 0.48, 0.473, 0.478, 0.485, 0.474, 0.494, 0.512, 0.485, 0.502]

PC2 loadings, middlegame (3.7% of variance), side to move at the bottom

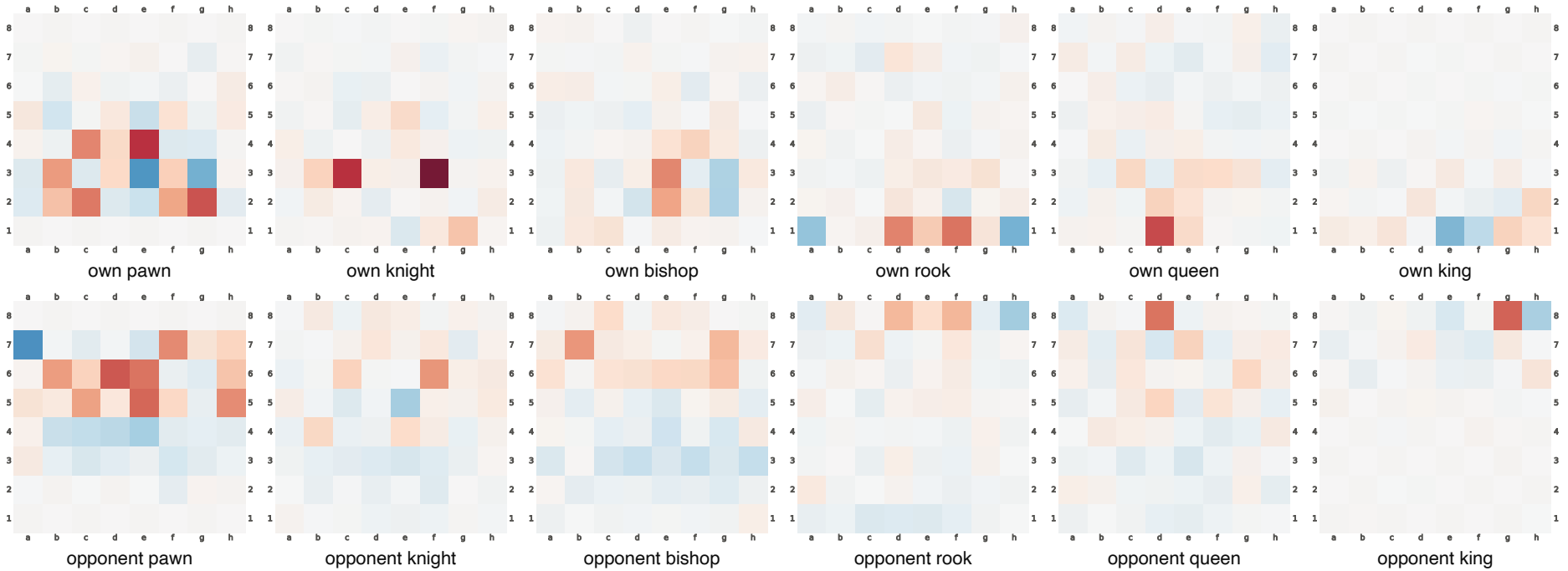


red = a piece on this square raises the PC score, blue = lowers it

-0.306 0.306

`middlegame PC2 failure rate by score decile (low -> high): [0.468, 0.484, 0.488, 0.48, 0.492, 0.494, 0.49, 0.492, 0.494, 0.491]`

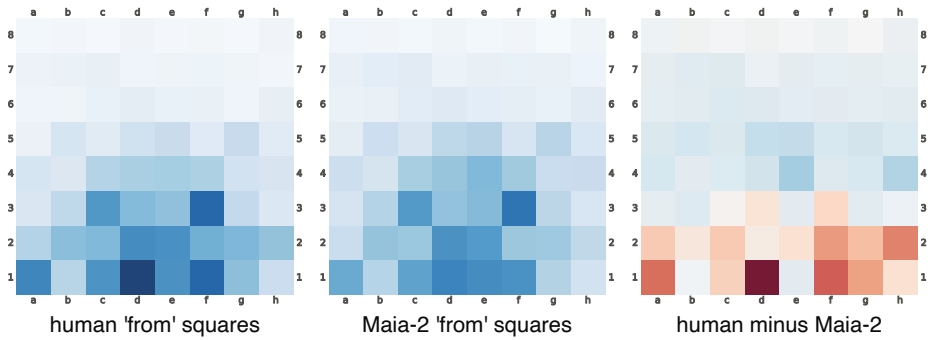
Where pieces stand when Maia-2 fails vs. succeeds (middlegame, n=20000)



percentage points: red = occupied more often in failed positions, blue = in correct ones

-1.73 1.73

Failed predictions, middlegame: 'from' squares (n=136014, mover at the bottom)



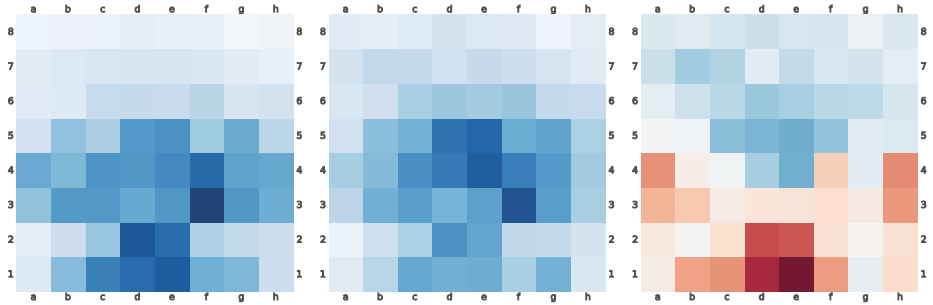
share of moves in %

0 6.05

red = humans use this square more, blue = Maia-2 expects it more (pp)

-1.67 1.67

Failed predictions, middlegame: 'to' squares (n=136014, mover at the bottom)



human 'to' squares | Maia-2 'to' squares | human minus Maia-2

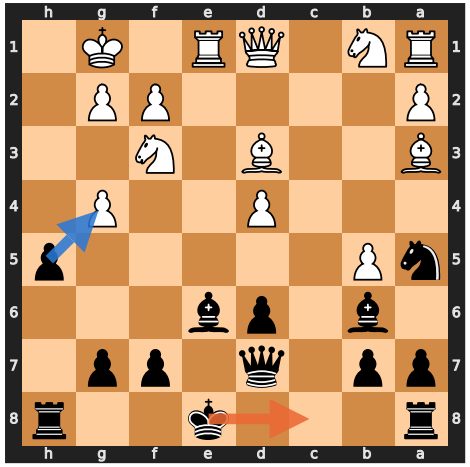
share of moves in %

0 4.11

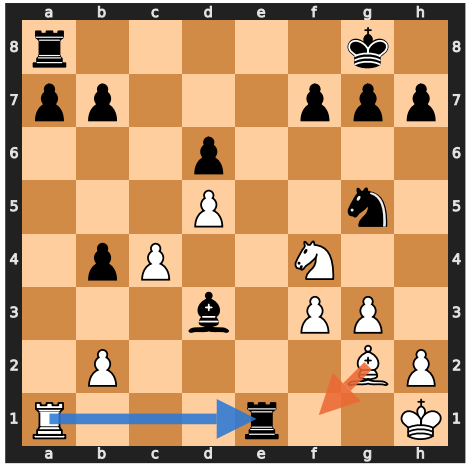
red = humans use this square more, blue = Maia-2 expects it more (pp)

-1.47 1.47

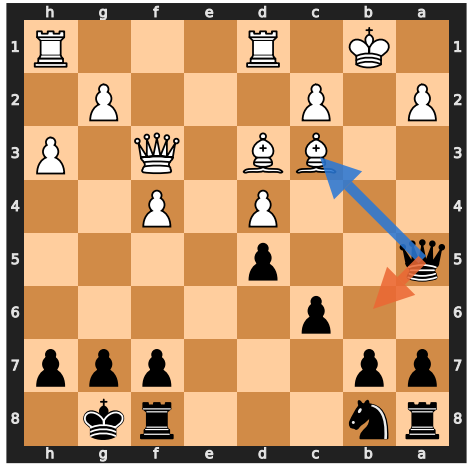
Most confident Maia-2 failures (middlegame)



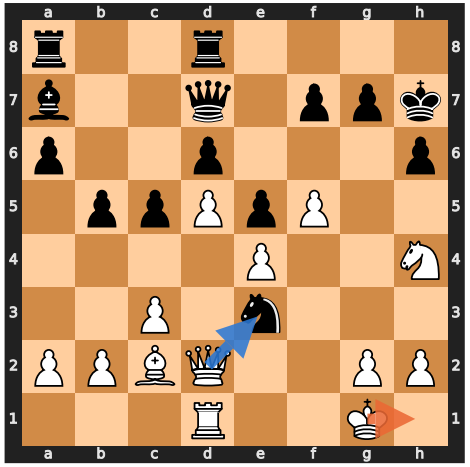
black to move, 2096 vs 2160, 109s left
Maia-2 h5g4 (100%) | human e8c8 (0%)
Italian Game: Classical Variation, Center Attack



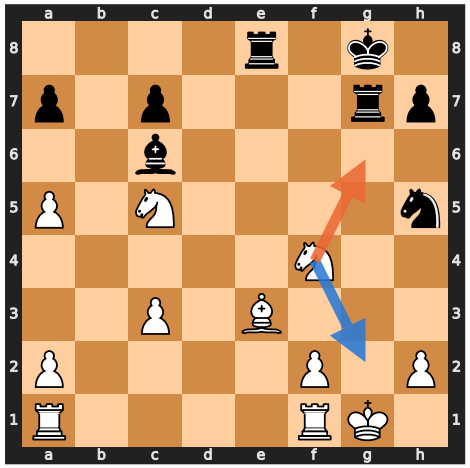
white to move, 1375 vs 941, 193s left
Maia-2 a1e1 (100%) | human g2f1 (0%)
English Opening: King's English Variation, Reversed Closed Sicilian



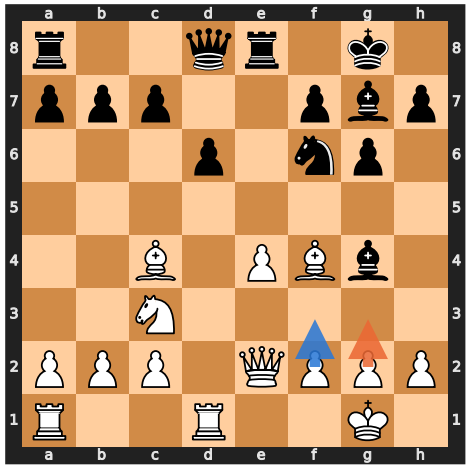
black to move, 1582 vs 1604, 133s left
Maia-2 a5c3 (100%) | human a5b6 (0%)
French Defense: Normal Variation



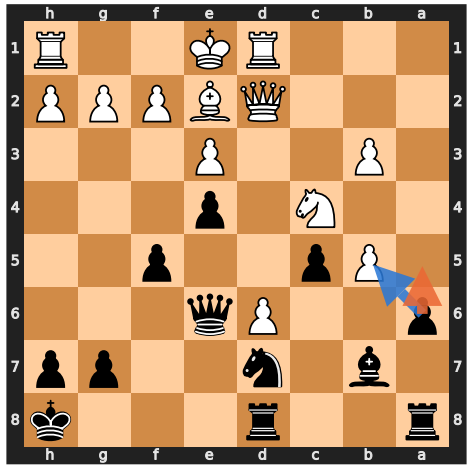
white to move, 1893 vs 1880, 7s left
Maia-2 d2e3 (100%) | human g1h1 (0%)
Ruy Lopez: Morphy Defense, Neo-Arkhangelsk Variation



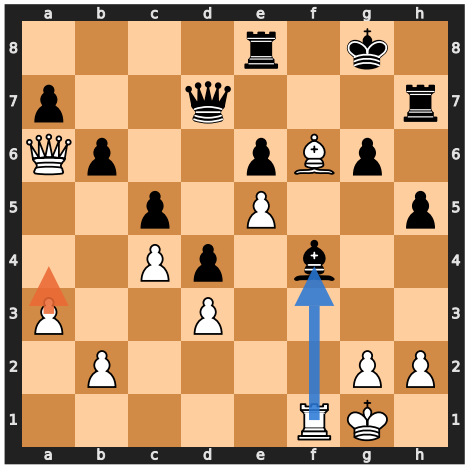
white to move, 1784 vs 2075, 125s left
Maia-2 f4g2 (100%) | human f4g6 (0%)
Ruy Lopez: Schliemann Defense, Exchange Variation



white to move, 2326 vs 2312, 161s left
Maia-2 f2f3 (100%) | human g2g3 (0%)
Pirc Defense: Classical Variation



black to move, 1642 vs 1673, 130s left
Maia-2 a6b5 (100%) | human a6a5 (0%)
Trompowsky Attack: Classical Defense



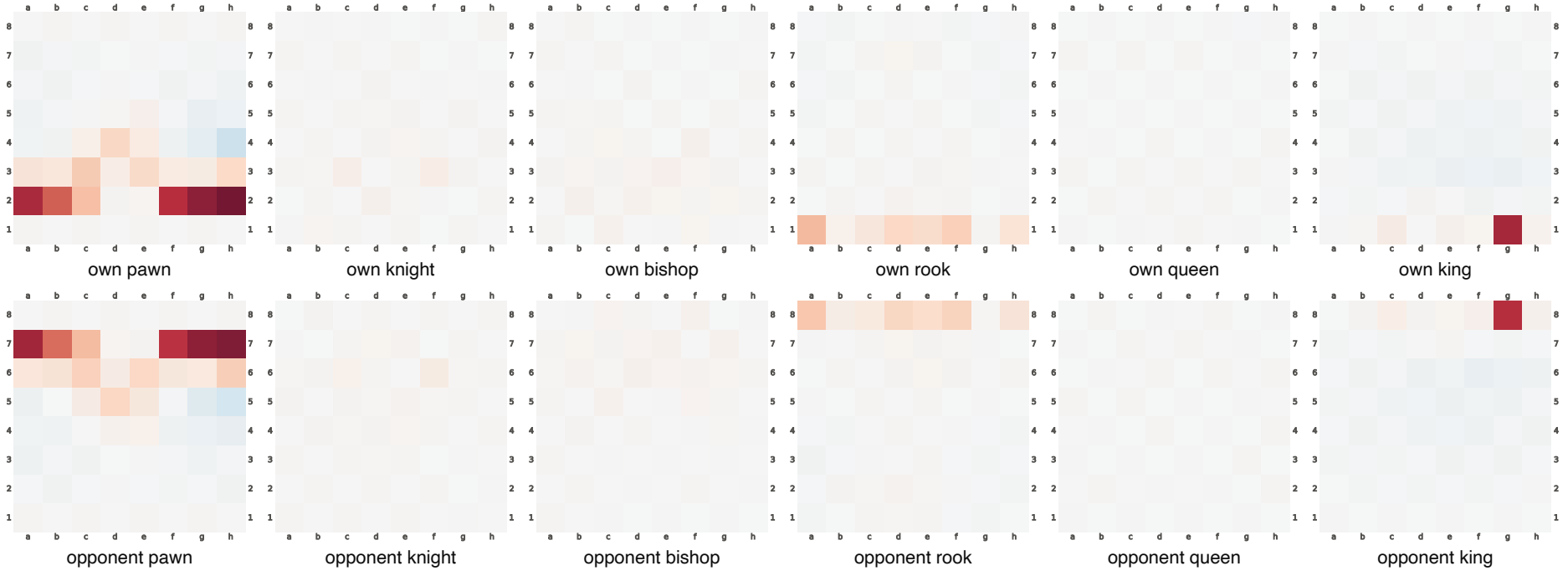
white to move, 1747 vs 1779, 109s left
Maia-2 f1f4 (100%) | human a3a4 (0%)
Bird Opening

blue arrow = Maia-2 top prediction, orange arrow = move actually played

Phase: endgame

endgame: 151193 non-forced positions | failure rate 0.447 | time_pressure mix {'>60s': 0.63, '<30s': 0.23, '30-60s': 0.14}

PC1 loadings, endgame (5.0% of variance), side to move at the bottom

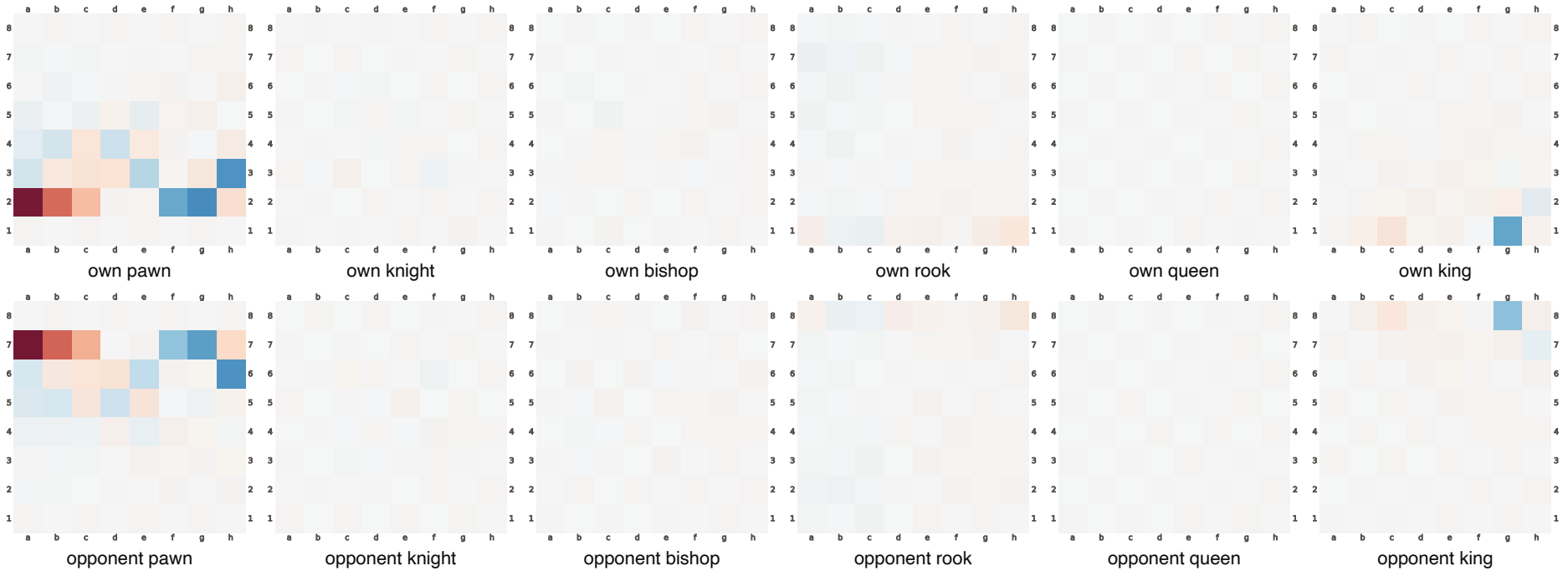


red = a piece on this square raises the PC score, blue = lowers it

-0.315 0.315

endgame PC1 failure rate by score decile (low -> high): [0.412, 0.394, 0.429, 0.468, 0.459, 0.453, 0.47, 0.473, 0.478, 0.472]

PC2 loadings, endgame (2.7% of variance), side to move at the bottom

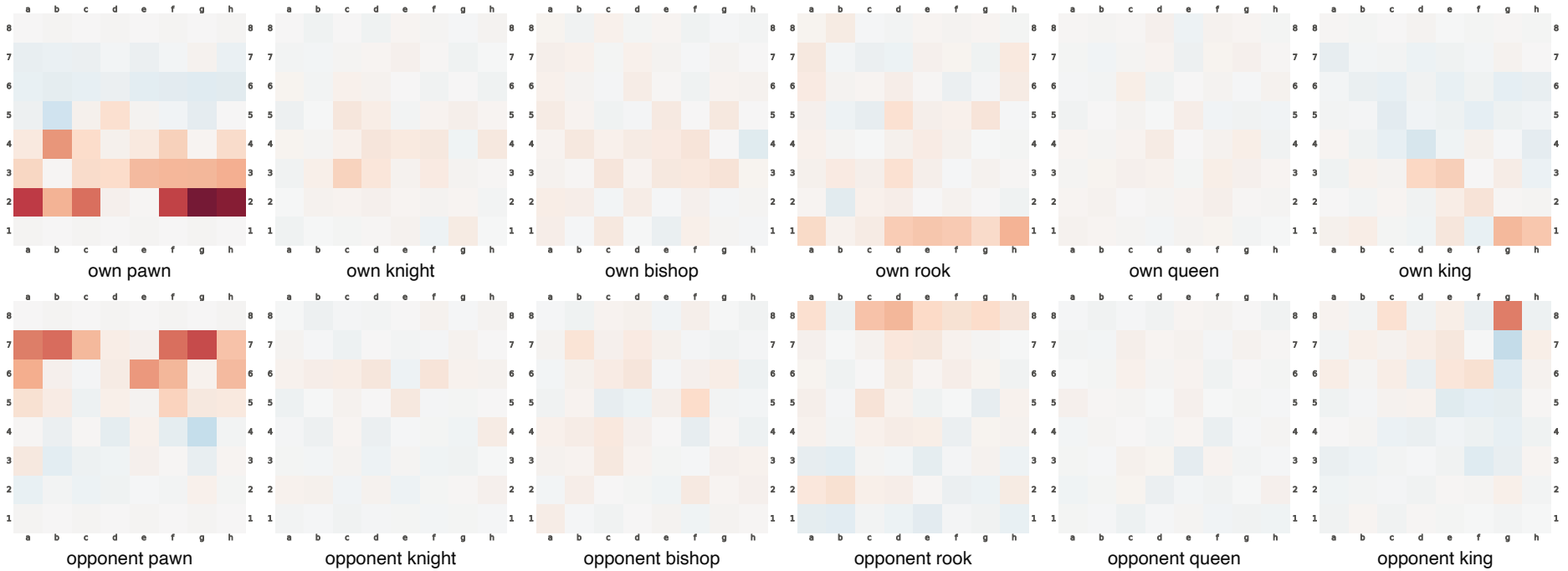


red = a piece on this square raises the PC score, blue = lowers it

-0.393 0.393

endgame PC2 failure rate by score decile (low -> high): [0.464, 0.472, 0.462, 0.429, 0.448, 0.438, 0.43, 0.452, 0.45, 0.464]

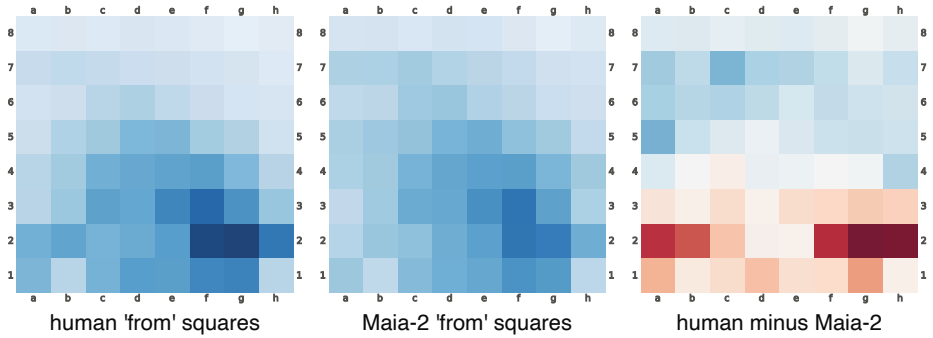
Where pieces stand when Maia-2 fails vs. succeeds (endgame, n=20000)



percentage points: red = occupied more often in failed positions, blue = in correct ones

-2.53 2.53

Failed predictions, endgame: 'from' squares (n=67582, mover at the bottom)



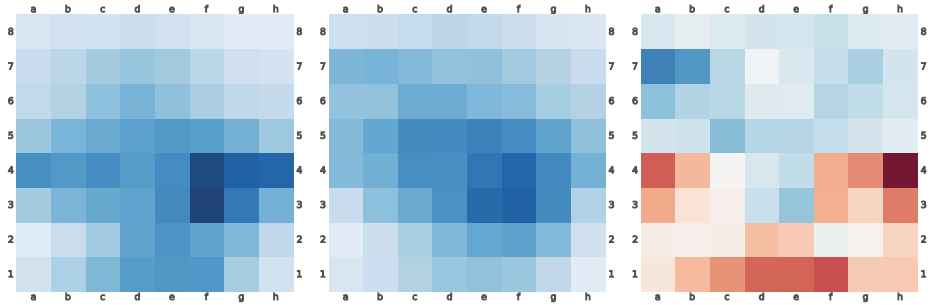
share of moves in %

0 3.9

red = humans use this square more, blue = Maia-2 expects it more (pp)

-0.954 0.954

Failed predictions, endgame: 'to' squares (n=67582, mover at the bottom)



human 'to' squares | Maia-2 'to' squares | human minus Maia-2

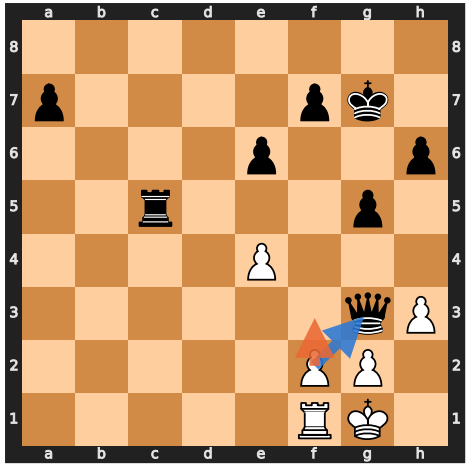
share of moves in %

0 3.48

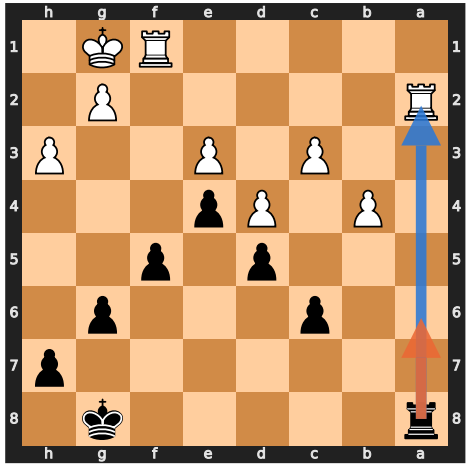
red = humans use this square more, blue = Maia-2 expects it more (pp)

-1.14 1.14

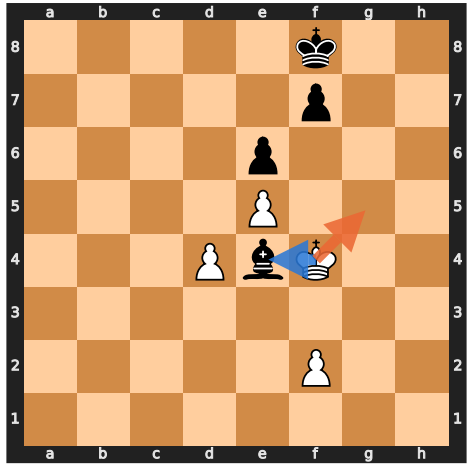
Most confident Maia-2 failures (endgame)



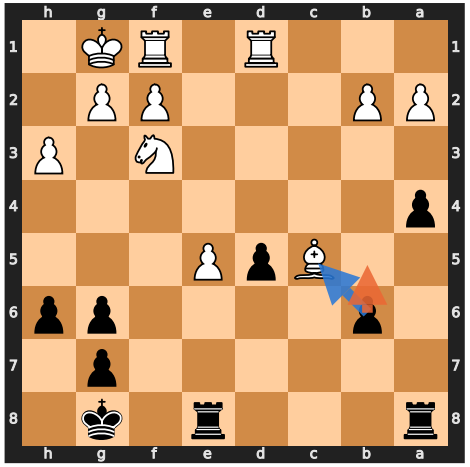
white to move, 2047 vs 2079, 183s left
Maia-2 f2g3 (100%) | human f2f3 (0%)
Pirc Defense



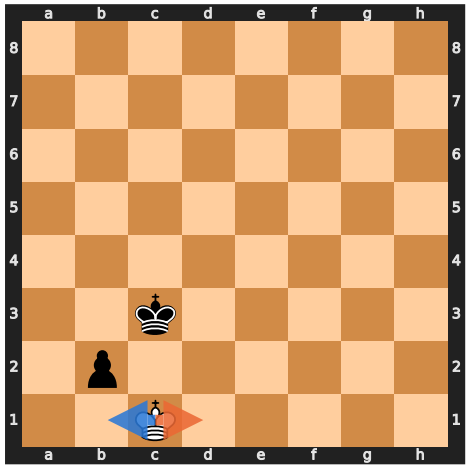
black to move, 1053 vs 1013, 404s left
Maia-2 a8a2 (100%) | human a8a6 (0%)
Queen's Pawn Game: Accelerated London System



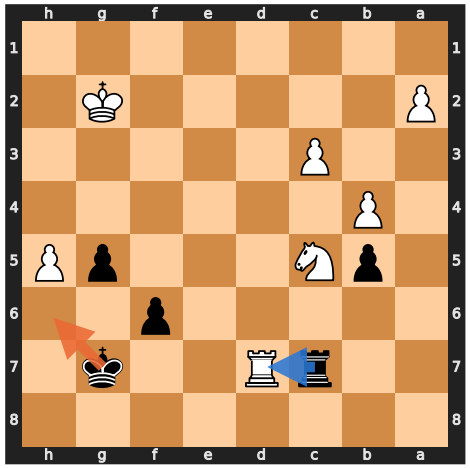
white to move, 2442 vs 2563, 4s left
Maia-2 f4e4 (100%) | human f4g5 (0%)
Grünfeld Defense: Exchange Variation, Spassky Variation



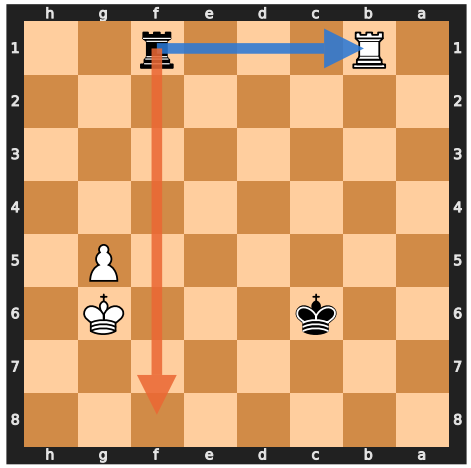
black to move, 1833 vs 1916, 147s left
Maia-2 b6c5 (100%) | human b6b5 (0%)
Caro-Kann Defense: Two Knights Attack



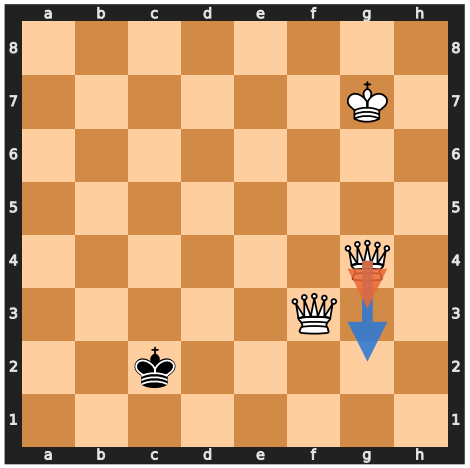
white to move, 1964 vs 2056, 140s left
Maia-2 c1b1 (100%) | human c1d1 (0%)
Queen's Pawn Game: Chigorin Variation



black to move, 1860 vs 1854, 50s left
Maia-2 c7d7 (100%) | human g7h6 (0%)
Caro-Kann Defense



black to move, 1565 vs 1563, 2s left
Maia-2 f1b1 (100%) | human f1f8 (0%)
Caro-Kann Defense: Hillbilly Attack



white to move, 1519 vs 1527, 1s left
Maia-2 g4g2 (100%) | human g4g3 (0%)
Sicilian Defense: Old Sicilian

blue arrow = Maia-2 top prediction, orange arrow = move actually played